

Identify Tables

Claims Table – core claim details:

Claim_ID Policy_ID Accident_Date FNOL_Date Claim_Type (collision, theft, fire) Estimated_Claim_Amount
Ultimate_Claim_Amount (final payout) Settlement_Date

Policyholder Table – customer and vehicle info:

Policy_ID & Customer_ID Driver_age Driving_experience Vehicle_type and Vehicle_age

Third-Party Involvement Table – accident severity:

Claim_ID (links to Claims) Number_of_third_parties Injury_severity (minor, serious, fatal)

Definitions

FNOL (First Notification of Loss)

The first report a customer makes to the insurer about an accident or loss. It contains partial information about the incident and is used to start the claims process.

Estimated Claim Amount

The initial cost estimate of a claim at the FNOL stage. Based on preliminary information, before full assessment.

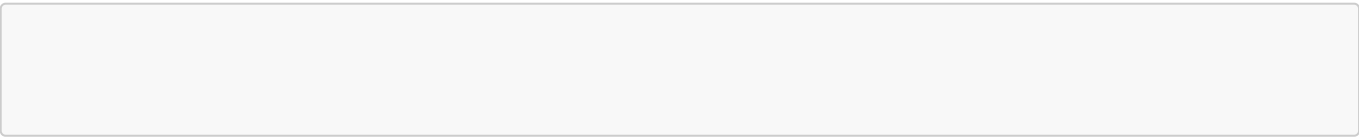
Ultimate Claim Amount

The final payout for the claim after investigation, repairs, legal costs, and third-party involvement are accounted for.

Third-Party Severity

Indicates the seriousness of injuries or damages to other parties involved in the accident: Minor → small or no medical attention Serious → significant injuries or hospitalization Fatal → death resulting from the accident

Sample claim cost histogram



01_kickoff_summary

Goal: Build a machine learning model to predict ultimate claim costs at FNOL stage.

Target: Accurate prediction of claim costs to improve reserve allocation and claims processing efficiency.

Why it matters:

Financial stability: prevent over/under-reserving

Operational efficiency: prioritize high-cost claims

Customer satisfaction: faster settlements and transparency

Metrics:

Reserve allocation accuracy improvement (%)

Prediction error metrics: MAE, RMSE, R^2

Operational metrics: average time to settle claims

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1. Goal:

To build a machine learning model that predicts the ultimate claim cost at the First Notification of Loss (FNOL) stage using historical claim, policy, and third-party data.

The model will provide real-time predictions to claims handlers, allowing for more accurate reserve allocation, faster settlements, and improved financial stability.

2. Target:

Predict Ultimate Claim Amount for each claim.

Identify the most influential FNOL and policy variables affecting claim cost.

Improve reserve allocation accuracy by at least 15% compared to current benchmarks.

3. Why it Matters:

Financial stability: accurate cost predictions ensure adequate but efficient reserves.

Operational performance: claims teams can prioritize high-cost or complex claims earlier.

Customer satisfaction: early and transparent settlements reduce disputes and waiting times.

Competitive advantage: predictive analytics is increasingly standard in leading insurers.

4. Metrics:

Prediction accuracy metrics:Mean Absolute Error (MAE),Root Mean Squared Error (RMSE),R² Score

Reserve allocation improvement: percentage improvement compared to current estimates

Operational metrics:Average time to settle claims,Number of high-cost claims flagged early

5. Data Tables:

Claims: (Core claim details) Claim_ID, Policy_ID, Accident_Date, FNOL_Date, Claim_Type, Estimated_Claim_Amount, Ultimate_Claim_Amount, Settlement_Date

Policyholder: (Customer & vehicle info) Policy_ID, Customer_ID, Driver_Age, Driving_Experience, Vehicle_Type, Vehicle_Age

Third-Party Involvement: (Accident severity) Claim_ID, Number_of_Third_Parties, Injury_Severity

7. Key Definitions:

FNOL (First Notification of Loss): Initial report of an accident/loss.

Estimated Claim Amount: Preliminary claim cost estimate at FNOL.

Ultimate Claim Amount: Final payout after full investigation.

Third-Party Severity: Degree of injury or damage to other parties (Minor, Serious, Fatal).